

DETAILED DESCRIPTION OF THE INVENTION — DIGITAL IDENTITY, PROFILE TOKENS, AND PHYSIOLOGIC SIGNATURE REPRESENTATIONS

The intraluminal physiologic monitoring system may generate digital identity constructs representing individualized physiologic characteristics derived from longitudinal sensing data. These digital identities may function as privacy-preserving representations of physiologic patterns without revealing explicit anatomical measurements.

In various embodiments, physiologic data may be transformed into profile tokens, symbolic identifiers, encoded signatures, or algorithmically generated representations reflecting unique physiologic behavior. Such identities may persist across sessions, devices, or platforms while remaining abstracted from raw sensor measurements.

Digital physiologic signatures may be used for authentication, personalization, compatibility analysis, or ecosystem participation without requiring disclosure of underlying sensitive information. Identity representations may therefore act as intermediaries between user data and external systems.

Profile tokens may be generated through machine learning models, statistical encoding techniques, hashing mechanisms, or multidimensional feature extraction processes capturing stable physiologic characteristics over time.

Certain embodiments may support portable identity frameworks allowing users to transport physiologic profiles between applications, services, or devices while maintaining ownership and permission control. Identity tokens may be stored locally, within secure cloud environments, or within decentralized identity systems.

Digital identity representations may enable matching systems, compatibility evaluation platforms, research participation frameworks, or personalized service environments operating without exposure of identifiable physiologic measurements.

Identity management systems may incorporate encryption, revocable credentials, token expiration mechanisms, or permission-based sharing protocols ensuring user control over how physiologic identity information is utilized.

Future embodiments may integrate blockchain-based identity verification, decentralized identifiers, or self-sovereign identity architectures enabling trusted verification without centralized ownership of personal data.

The digital identity and physiologic signature systems described herein are illustrative rather than limiting. Any representation capable of encoding individualized physiologic information into portable, privacy-preserving identity constructs is considered within the scope of the present disclosure.